

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Chapter 6: The Riemann–Stieltjes Integral

Section: Integration of Vector-Valued Functions

Based on Walter Rudin's *Principles of Mathematical Analysis*

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Outline

$$\int_a^b f(x) dg(x)$$

$$S(x) = \sum_{i=1}^n f(x_i^*)(g(x_i) - g(x_{i-1}))) = S(x_{n-1}, x_n)$$

Motivation

Integrating Componentwise

The Fundamental Theorem, Vector Version

The Triangle Inequality for Integrals

Summary



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

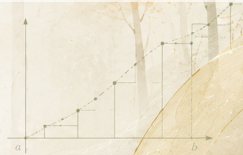
Motivation



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

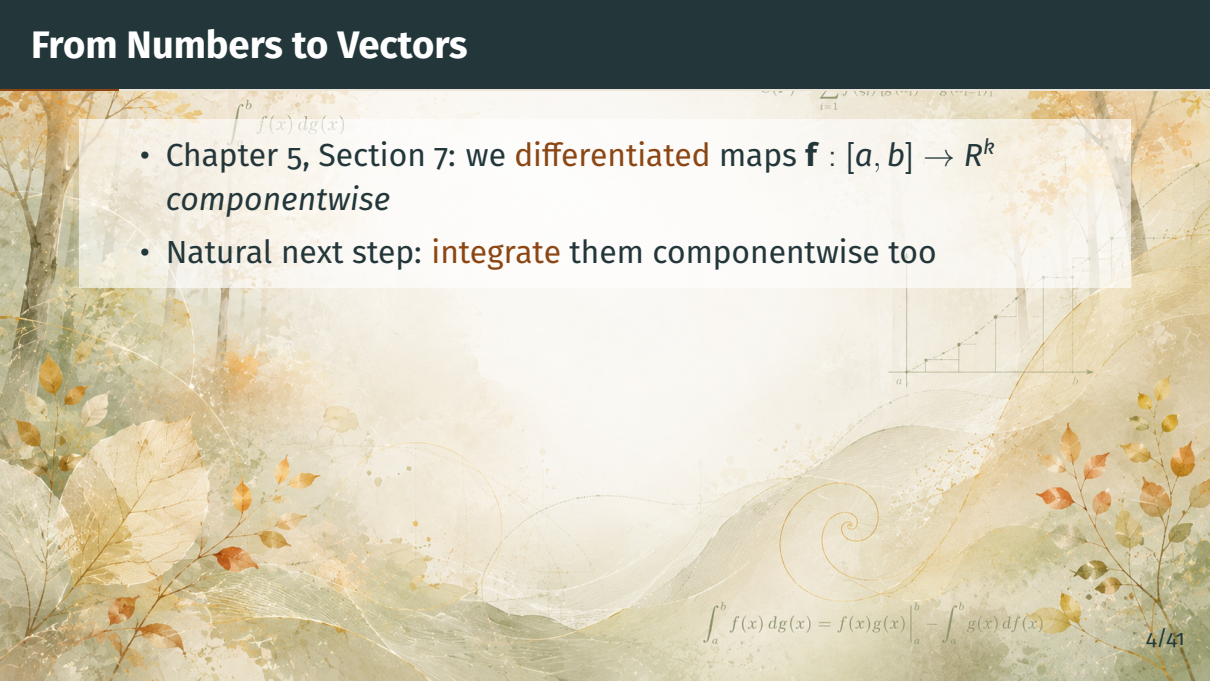
From Numbers to Vectors

- Chapter 5, Section 7: we **differentiated** maps $\mathbf{f} : [a, b] \rightarrow \mathbb{R}^k$ *componentwise*



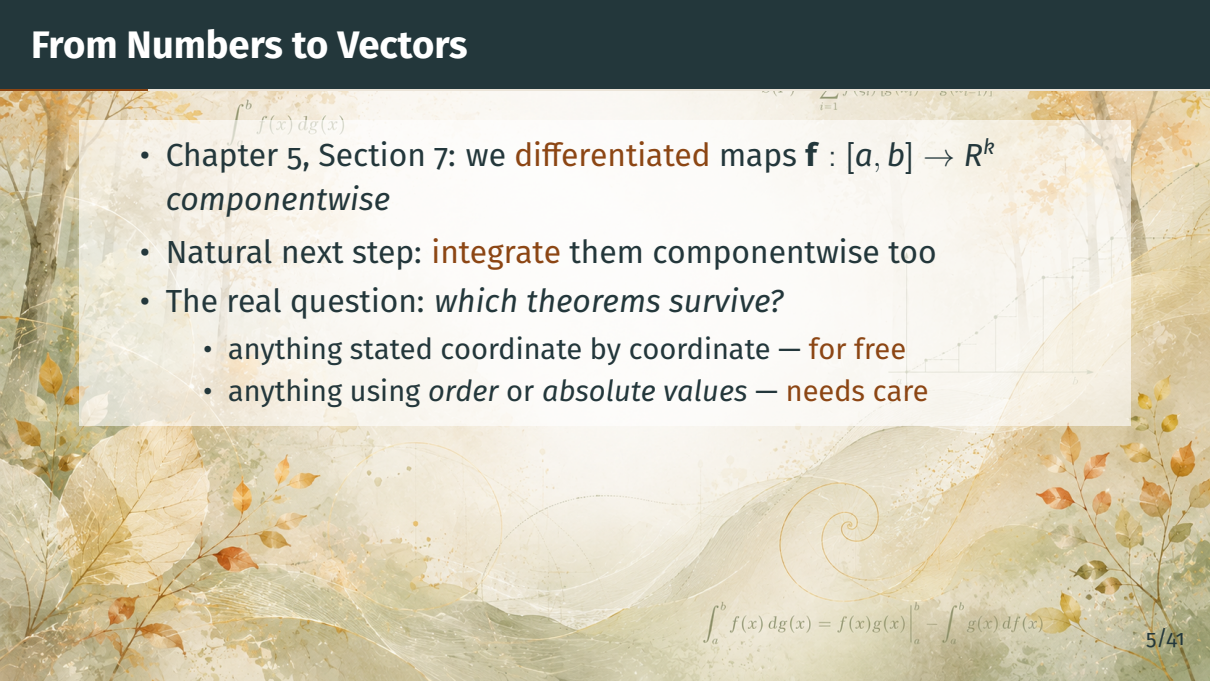
$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

From Numbers to Vectors

- 
- $\int_a^b f(x) dg(x)$
- Chapter 5, Section 7: we **differentiated** maps $\mathbf{f} : [a, b] \rightarrow \mathbb{R}^k$ *componentwise*
 - Natural next step: **integrate** them componentwise too

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

From Numbers to Vectors

- 
- The background features a soft-focus illustration of a forest with trees and autumn leaves in shades of orange, yellow, and green. Overlaid on this is a semi-transparent white box containing text. In the upper right corner, there is a small graph showing a step function and a smooth curve, with axes labeled 'a' and 'b'. At the top of the white box, the integral $\int_a^b f(x) dg(x)$ is written. The main content consists of a bulleted list.
- Chapter 5, Section 7: we **differentiated** maps $\mathbf{f} : [a, b] \rightarrow \mathbb{R}^k$ *componentwise*
 - Natural next step: **integrate** them componentwise too
 - The real question: *which theorems survive?*
 - anything stated coordinate by coordinate — **for free**
 - anything using *order* or *absolute values* — **needs care**

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

The Picture to Keep in Mind

$$\int_a^b f(x) dg(x)$$

path $\mathbf{F}(t)$

$$\mathbf{F}(b) - \mathbf{F}(a)$$

$$= \int_a^b \mathbf{f} dt$$

$\mathbf{F}(b)$

$\mathbf{F}(a)$

$$\mathbf{f}(t) = \mathbf{F}'(t)$$

Displacement $\left| \int_a^b \mathbf{f} dt \right|$ versus **distance traveled** $\int_a^b |\mathbf{f}| dt$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

What We Will Prove

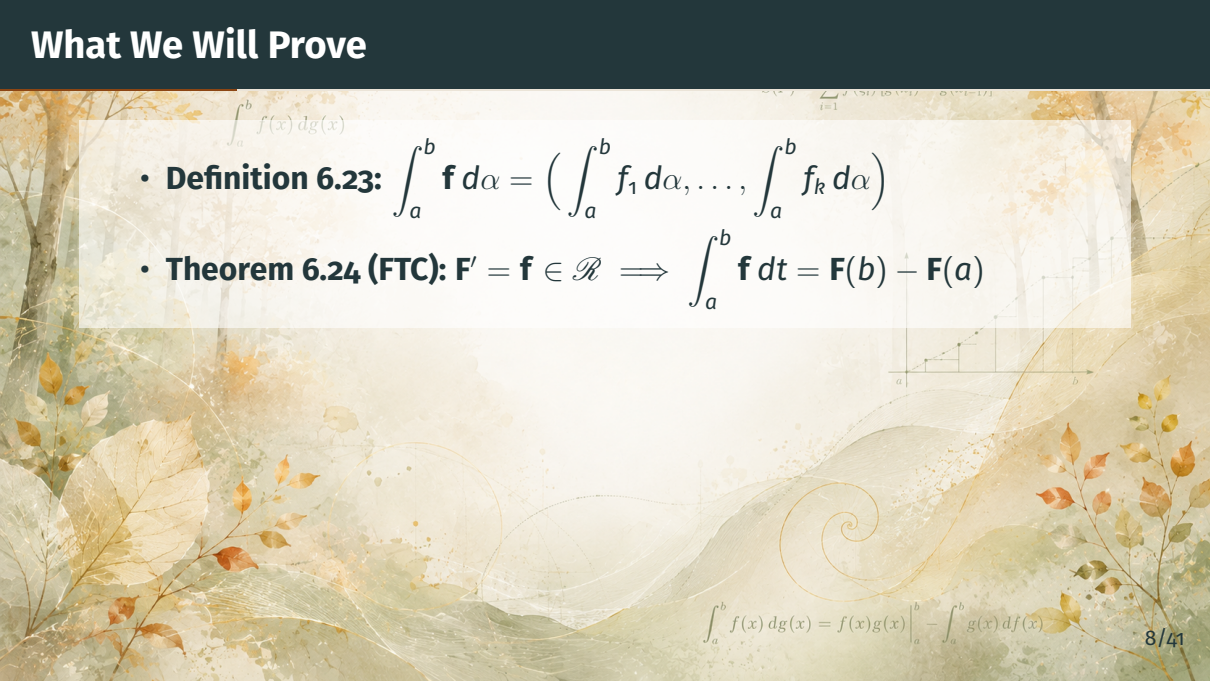
$\int_a^b f(x) dg(x)$

- **Definition 6.23:** $\int_a^b \mathbf{f} d\alpha = \left(\int_a^b f_1 d\alpha, \dots, \int_a^b f_k d\alpha \right)$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

What We Will Prove

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- **Definition 6.23:** $\int_a^b \mathbf{f} d\alpha = \left(\int_a^b f_1 d\alpha, \dots, \int_a^b f_k d\alpha \right)$
- **Theorem 6.24 (FTC):** $F' = \mathbf{f} \in \mathcal{R} \implies \int_a^b \mathbf{f} dt = \mathbf{F}(b) - \mathbf{F}(a)$

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What We Will Prove

- **Definition 6.23:** $\int_a^b \mathbf{f} d\alpha = \left(\int_a^b f_1 d\alpha, \dots, \int_a^b f_k d\alpha \right)$
- **Theorem 6.24 (FTC):** $F' = \mathbf{f} \in \mathcal{R} \implies \int_a^b \mathbf{f} dt = \mathbf{F}(b) - \mathbf{F}(a)$
- **Theorem 6.25:** $\mathbf{f} \in \mathcal{R}(\alpha) \implies |\mathbf{f}| \in \mathcal{R}(\alpha)$ and $\left| \int_a^b \mathbf{f} d\alpha \right| \leq \int_a^b |\mathbf{f}| d\alpha$
new ingredient: the Schwarz inequality

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Integrating Componentwise



Definition 6.23: Vector-Valued Integrals

Definition 6.23

Let $f_1, \dots, f_k$ be real functions on $[a, b]$, and let $\mathbf{f} = (f_1, \dots, f_k)$ be the corresponding mapping of $[a, b]$ into R^k . If α increases monotonically on $[a, b]$, to say that $\mathbf{f} \in \mathcal{R}(\alpha)$ means that $f_j \in \mathcal{R}(\alpha)$ for $j = 1, \dots, k$.

Definition 6.23: Vector-Valued Integrals

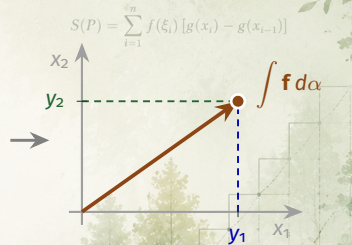
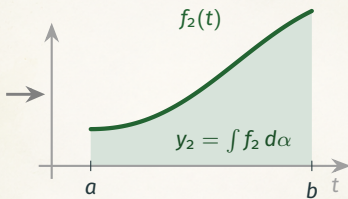
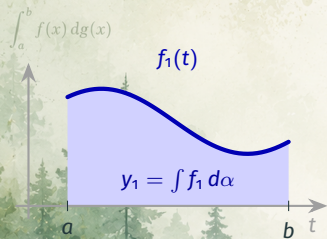
Definition 6.23

Let $f_1, \dots, f_k$ be real functions on $[a, b]$, and let $\mathbf{f} = (f_1, \dots, f_k)$ be the corresponding mapping of $[a, b]$ into $\mathbb{R}^k$. If α increases monotonically on $[a, b]$, to say that $\mathbf{f} \in \mathcal{R}(\alpha)$ means that $f_j \in \mathcal{R}(\alpha)$ for $j = 1, \dots, k$. If this is the case, we define

$$\int_a^b \mathbf{f} d\alpha = \left(\int_a^b f_1 d\alpha, \dots, \int_a^b f_k d\alpha \right).$$

$\int \mathbf{f} d\alpha$ is the **point in $\mathbb{R}^k$** whose j th coordinate is $\int f_j d\alpha$.

The Definition in a Picture ($k = 2$)



Recipe: integrate each component separately, then assemble the results into one point of R^k .

What Carries Over for Free

$\int_a^b f(x) dg(x)$
Apply the real-valued result to each coordinate:

- **Theorem 6.12(a): linearity:** $\int (\mathbf{f} + \mathbf{g}) d\alpha = \int \mathbf{f} d\alpha + \int \mathbf{g} d\alpha$,
 $\int c\mathbf{f} d\alpha = c \int \mathbf{f} d\alpha$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$a = x_0 \quad x_1 \quad x_2 \quad \dots \quad x_{n-1} \quad x_n = b$$

What Carries Over for Free

Apply the real-valued result to each coordinate:

- **Theorem 6.12(a):** *linearity:* $\int (\mathbf{f} + \mathbf{g}) d\alpha = \int \mathbf{f} d\alpha + \int \mathbf{g} d\alpha$,
 $\int c\mathbf{f} d\alpha = c \int \mathbf{f} d\alpha$
- **Theorem 6.12(c):** *splitting the interval:* $\int_a^c \mathbf{f} d\alpha + \int_c^b \mathbf{f} d\alpha = \int_a^b \mathbf{f} d\alpha$
- **Theorem 6.12(e):** *linearity in α :* $\int \mathbf{f} d(\alpha_1 + \alpha_2) = \int \mathbf{f} d\alpha_1 + \int \mathbf{f} d\alpha_2$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$$

What Carries Over for Free

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

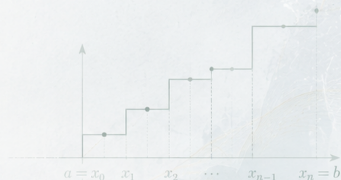
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- **Theorem 6.12(a):** *linearity:* $\int (\mathbf{f} + \mathbf{g}) d\alpha = \int \mathbf{f} d\alpha + \int \mathbf{g} d\alpha$,
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- **Theorems 6.17, 6.20, 6.21:** reduction to the Riemann integral
 $\int \mathbf{f} d\alpha = \int \mathbf{f}\alpha' dx$; continuity and differentiability of $\mathbf{F}(x) = \int_a^x \mathbf{f} dt$;
the **Fundamental Theorem**

$$\int_a^b f(x) dg(x)$$

The Fundamental Theorem, Vector Version

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

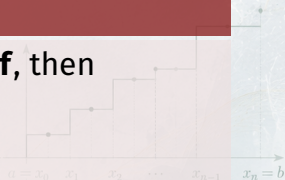
Theorem 6.24: The Fundamental Theorem of Calculus

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.24

If $\mathbf{f}$ and $\mathbf{F}$ map $[a, b]$ into $\mathbb{R}^k$, if $\mathbf{f} \in \mathcal{R}$ on $[a, b]$, and if $\mathbf{F}' = \mathbf{f}$, then

$$\int_a^b \mathbf{f}(t) dt = \mathbf{F}(b) - \mathbf{F}(a).$$



Interpretation: $\mathbf{f}$ = velocity, $\mathbf{F}$ = position. Integrating the velocity gives the **net displacement**.

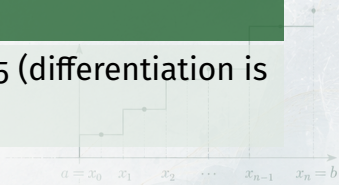
$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.24: Coordinate by Coordinate

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Step 1: Unpack the hypothesis

Write $\mathbf{f} = (f_1, \dots, f_k)$ and $\mathbf{F} = (F_1, \dots, F_k)$. By Chapter 5 (differentiation is componentwise), $\mathbf{F}' = \mathbf{f} \iff F'_j = f_j$ for $j = 1, \dots, k$.


$$a = x_0 \quad x_1 \quad x_2 \quad \dots \quad x_{n-1} \quad x_n = b$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.24: Coordinate by Coordinate

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

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Step 2: Apply the real FTC (Theorem 6.21) to each coordinate

Each $f_j \in \mathcal{R}$ and $F'_j = f_j$, so $\int_a^b f_j(t) dt = F_j(b) - F_j(a)$ for $j = 1, \dots, k$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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Step 3: Reassemble

Both sides of $\int_a^b \mathbf{f} dt = \mathbf{F}(b) - \mathbf{F}(a)$ have j th coordinate $F_j(b) - F_j(a)$. ■

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

A Worked Example: Half a Turn

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Integrating a rotating velocity

Let $\mathbf{f}(t) = (\cos t, \sin t)$ on $[0, \pi]$. A position function with $\mathbf{F}' = \mathbf{f}$ is

$$\mathbf{F}(t) = (\sin t, -\cos t).$$

By Theorem 6.24,

$$\int_0^\pi \mathbf{f}(t) dt = \mathbf{F}(\pi) - \mathbf{F}(0) = (0, 1) - (0, -1) = (0, 2).$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

A Worked Example: Half a Turn

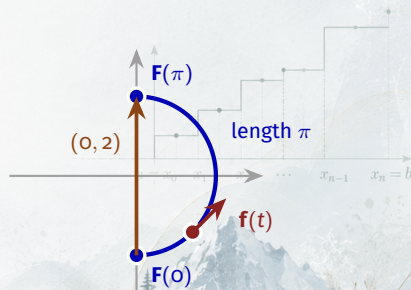
Integrating a rotating velocity

$$\mathbf{f}(t) = (\cos t, \sin t), \quad \mathbf{F}(t) = (\sin t, -\cos t)$$

$$\int_0^\pi \mathbf{f} dt = \mathbf{F}(\pi) - \mathbf{F}(0) = (0, 2)$$

$$|\mathbf{f}(t)| = 1, \text{ so } \int_0^\pi |\mathbf{f}| dt = \pi$$

$$\left| \int_0^\pi \mathbf{f} dt \right| = 2 < \pi = \int_0^\pi |\mathbf{f}| dt$$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

The Triangle Inequality for Integrals



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Why This One Needs a New Idea

Recall: Theorem 6.13(b), real case

If $f \in \mathcal{R}(\alpha)$, then $|f| \in \mathcal{R}(\alpha)$ and $\left| \int_a^b f d\alpha \right| \leq \int_a^b |f| d\alpha$.

The old proof: choose $c = \pm 1$ so that $c \int f d\alpha \geq 0$; then

$$\left| \int f d\alpha \right| = c \int f d\alpha = \int cf d\alpha \leq \int |f| d\alpha \quad (\text{since } cf \leq |f|).$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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For vectors:

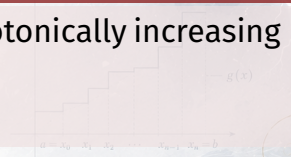
- $\int \mathbf{f} d\alpha$ has **no sign** — it is a point of R^k
- $|\mathbf{f}| = (f_1^2 + \cdots + f_k^2)^{1/2}$ is **not** built coordinate by coordinate

Theorem 6.25: The Triangle Inequality for Integrals

Theorem 6.25

If $\mathbf{f}$ maps $[a, b]$ into R^k and if $\mathbf{f} \in \mathcal{R}(\alpha)$ for some monotonically increasing function α on $[a, b]$, then

$$|\mathbf{f}| \in \mathcal{R}(\alpha).$$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.25: The Triangle Inequality for Integrals

Theorem 6.25

If $\mathbf{f}$ maps $[a, b]$ into R^k and if $\mathbf{f} \in \mathcal{R}(\alpha)$ for some monotonically increasing function α on $[a, b]$, then $|\mathbf{f}| \in \mathcal{R}(\alpha)$, and

$$\left| \int_a^b \mathbf{f} d\alpha \right| \leq \int_a^b |\mathbf{f}| d\alpha. \quad (40)$$

Why “triangle inequality”: for a finite sum, $|\mathbf{v}_1 + \cdots + \mathbf{v}_n| \leq |\mathbf{v}_1| + \cdots + |\mathbf{v}_n|$; an integral is a limit of such sums.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof Strategy: Theorem 6.25

$$\int_a^b f(x) dg(x)$$

Part 1: $|f| \in \mathcal{R}(\alpha)$.

- $|f| = \varphi(f_1^2 + \dots + f_k^2)$ with $\varphi = \sqrt{\cdot} \Rightarrow$ use the *composition theorem* (6.11) twice



Part 2: the inequality (40).

- Let $\mathbf{y} = \int \mathbf{f} d\alpha$ be the *answer*
- **Key trick:** measure $\mathbf{f}$ along the direction of $\mathbf{y}$: $|\mathbf{y}|^2 = \int \mathbf{y} \cdot \mathbf{f} d\alpha$
- **Schwarz:** $\mathbf{y} \cdot \mathbf{f}(t) \leq |\mathbf{y}| |\mathbf{f}(t)| \Rightarrow |\mathbf{y}|^2 \leq |\mathbf{y}| \int |\mathbf{f}| d\alpha$
- Divide by $|\mathbf{y}|$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.25, Part 1: $|f|$ Is Integrable

$$\int_a^b f(x) dg(x)$$

Step 1: Write $|f|$ in terms of the components

If $f_1, \dots, f_k$ are the components of f , then

$$S(P) = \sum_{i=1}^k f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$|f| = (f_1^2 + \dots + f_k^2)^{1/2}.$$

(41)



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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$$|f| = (f_1^2 + \dots + f_k^2)^{1/2}. \quad (41)$$



Step 2: The inside is integrable

By Theorem 6.11 (with $\varphi(t) = t^2$), each $f_i^2 \in \mathcal{R}(\alpha)$; hence by Theorem 6.12(a) so does their sum $g = f_1^2 + \dots + f_k^2 \in \mathcal{R}(\alpha)$, with $0 \leq g \leq M$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.25, Part 1: $|f|$ Is Integrable

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Step 3: The square root is continuous

$x \mapsto x^2$ is a continuous 1-1 map of the compact set $[0, \sqrt{M}]$ onto $[0, M]$, so by Theorem 4.17 its inverse $\sqrt{\cdot}$ is continuous on $[0, M]$ (any $M > 0$).

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

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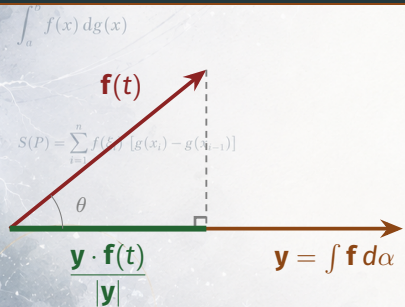
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Step 4: Compose once more

Theorem 6.11 with $\varphi = \sqrt{\cdot}$ and (41) give $|f| = \sqrt{g} \in \mathcal{R}(\alpha)$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

The Key Idea: Measure f in the Direction of the Answer



Schwarz (Theorem 1.35):

$$\mathbf{y} \cdot \mathbf{f}(t) = |\mathbf{y}| |\mathbf{f}(t)| \cos \theta \leq |\mathbf{y}| |\mathbf{f}(t)|$$

the shadow of $\mathbf{f}(t)$ on $\mathbf{y}$

is never longer than $\mathbf{f}(t)$ itself

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.25, Part 2: The Inequality (40)

$$\int_a^b f(x) dg(x)$$

Step 1: Name the answer

Put $\mathbf{y} = (y_1, \dots, y_k)$ with $y_j = \int f_j d\alpha$, so $\mathbf{y} = \int \mathbf{f} d\alpha$. **Goal:** $|\mathbf{y}| \leq \int |\mathbf{f}| d\alpha$.

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.25, Part 2: The Inequality (40)

Step 1: Name the answer

Put $\mathbf{y} = (y_1, \dots, y_k)$ with $y_j = \int f_j d\alpha$, so $\mathbf{y} = \int \mathbf{f} d\alpha$. **Goal:** $|\mathbf{y}| \leq \int |\mathbf{f}| d\alpha$.

Step 2: Express $|\mathbf{y}|^2$ as an integral (linearity, Theorem 6.12(a))

$$|\mathbf{y}|^2 = \sum_j y_j^2 = \sum_j y_j \int f_j d\alpha = \int \left(\sum_j y_j f_j \right) d\alpha.$$

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Step 4: Integrate (42) — monotonicity, Theorem 6.12(b)

Both sides of (42) lie in $\mathcal{R}(\alpha)$ (Part 1 for the right side), so

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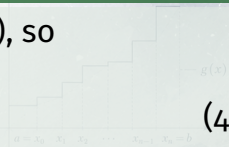
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Step 5: Divide by $|\mathbf{y}|$

If $\mathbf{y} = \mathbf{0}$, (40) is trivial. If $\mathbf{y} \neq \mathbf{0}$, divide (43) by $|\mathbf{y}| > 0$:

$$|\mathbf{y}| \leq \int |\mathbf{f}| d\alpha,$$

which is (40). ■

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Remarks: What Theorem 6.25 Says

$$\int_a^b f(x) dg(x)$$

Geometric meaning

With $\mathbf{f} = \mathbf{F}'$ and $\alpha(t) = t$, Theorem 6.24 turns (40) into

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$\underbrace{|\mathbf{F}(b) - \mathbf{F}(a)|}_{\text{displacement}} \leq \underbrace{\int_a^b |\mathbf{F}'(t)| dt}_{\text{distance traveled}}.$$



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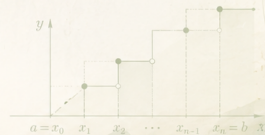
Two more consequences

- **Theorem 6.12(d) for vectors:** $|\mathbf{f}| \leq M \Rightarrow \left| \int_a^b \mathbf{f} d\alpha \right| \leq M[\alpha(b) - \alpha(a)]$
- **Theorem 5.19** (mean value inequality): $|\mathbf{F}(b) - \mathbf{F}(a)| \leq (b - a) |\mathbf{F}'(x)|$ for some x (same idea, no integral)

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

Summary



$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Summary

Key takeaways from this section:

- **Definition 6.23:** integrate **componentwise**; linearity, interval splitting, Theorems 6.17, 6.20, 6.21 all carry over *for free*

$$\int_a^b f(x) dg(x)$$



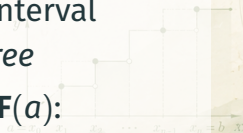
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- **Theorem 6.24 (FTC):** $\mathbf{F}' = \mathbf{f} \in \mathcal{R} \implies \int_a^b \mathbf{f} \, dt = \mathbf{F}(b) - \mathbf{F}(a)$:
integrated velocity is displacement

$$\int_a^b f(x) dg(x)$$



$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Summary

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- **Definition 6.23:** integrate **componentwise**; linearity, interval splitting, Theorems 6.17, 6.20, 6.21 all carry over *for free*
- **Theorem 6.24 (FTC):** $\mathbf{F}' = \mathbf{f} \in \mathcal{R} \implies \int_a^b \mathbf{f} \, dt = \mathbf{F}(b) - \mathbf{F}(a)$: *integrated velocity is displacement*
- **Theorem 6.25:** $|\mathbf{f}| \in \mathcal{R}(\alpha)$ and $|\int \mathbf{f} \, d\alpha| \leq \int |\mathbf{f}| \, d\alpha$: *displacement $\leq$ distance traveled*; proof: dot with the answer $\mathbf{y}$, then **Schwarz**

Coming up next: **rectifiable curves** — the length of a curve, and why it equals $\int_a^b |\gamma'(t)| \, dt$.